

AUTOMATION OF THERMAL POWER PLANT USING PLC LADDER LOGIC

A report submitted in partial fulfillment of the requirements

Of

Programming Logic

Control-Project (EI62)

In

VIth Semester

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Abstract

This report presents the design and implementation of an automated control system for a Thermal Power Plant using Programmable Logic Controller (PLC) Ladder Logic. The project demonstrates sequential automation of major sub-systems including the ID fan, boiler-heater, steam valve, turbine, generator, cooling pump, conveyor, and ash pump. The control logic is designed using On-Delay Timers (TON) to ensure safe, sequential startup and shutdown of each component. The PLC addresses are mapped using Allen-Bradley Local I/O conventions with clearly defined input and output tags.

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1. Introduction

A Thermal Power Plant converts heat energy — primarily obtained from burning coal, natural gas, or other fuels — into electrical energy. The operation of such a plant involves numerous interdependent processes that must be started and stopped in a strict sequence to ensure safety and operational efficiency. Manual control of these processes is error-prone and potentially hazardous, making automation a critical requirement.

Programmable Logic Controllers (PLCs) are industrial digital computers widely used for automation of electromechanical processes. Ladder Logic, a graphical programming language derived from relay-based control circuits, is the most common language used to program PLCs. It allows engineers to represent control logic in a familiar, easy-to-debug format.

This project implements a PLC Ladder Logic program that automates the sequential startup and shutdown of all major sub-systems of a Thermal Power Plant, ensuring each component is activated only after the previous one has stabilised — using On-Delay Timers (TON) with a preset time of 5000 ms (5 seconds) for each stage.

2. System Overview

The automated Thermal Power Plant control system consists of a sequential chain of 10 major components. On pressing the Start button, the PLC energises each component one by one, with a 5-second delay between successive stages. A Stop button de-energises the entire system. A latching (seal-in) circuit ensures the system remains ON even after the Start button is released.

The sequence of operation is as follows:

Stage	Component	Output Tag	Trigger Condition
1	ID Fan (id_dan)	Local:2:O.Data.0	Start pressed + CN ON
2	Boiler Timer (t1)	Timer t1 – TON 5s	id_dan ON + Local:2:O.Data.0
3	Boiler-Heater	Local:2:O.Data.3	t1.DN + Local:2:O.Data.1
4	Burner (bunner)	Local:2:O.Data.2	t2.DN
5	Burner Timer (t3)	Timer t3 – TON 5s	bunner ON + Local:1:O.Data.2
6	Boiler Heater	Local:2:O.Data.3	t3.DN
7	Steam Valve	Local:2:O.Data.4	t4.DN
8	Turbine	Local:2:O.Data.5	t5.DN
9	Generator	Local:2:O.Data.6	t6.DN
10	Cooling Pump	Local:2:O.Data.7	t7.DN + Generator ON
11	Conveyor	Local:2:O.Data.0 (slot2)	t8.DN
12	Ash Pump	Local:2:O.Data.3 (slot2)	t9.DN + conveyor ON

3. Components of a Thermal Power Plant

The following are the major sub-systems automated in this project:

3.1 ID Fan (Induced Draft Fan) — id_dan

The Induced Draft Fan creates a negative pressure inside the boiler furnace, drawing flue gases out through the chimney. It is the first component to start in the sequence, as it is needed to establish airflow before combustion begins.

3.2 Boiler-Heater

The boiler converts water into high-pressure steam using heat from the combustion of fuel. The boiler-heater element is energised after the ID fan and burner are operational and sufficient airflow has been established.

3.3 Burner (bunner)

The burner ignites and sustains the combustion of fuel (coal, gas, or oil) inside the furnace. It is started after the ID fan has been running for 5 seconds to ensure safe combustion conditions.

3.4 Steam Valve

The steam valve controls the flow of steam from the boiler to the turbine. It opens only after the boiler has built up sufficient steam pressure, as indicated by the completion of the boiler-heater timer.

3.5 Turbine

High-pressure steam from the boiler passes through the turbine blades, converting thermal energy into mechanical rotational energy. The turbine starts after the steam valve has been open for 5 seconds.

3.6 Generator

The generator is mechanically coupled to the turbine shaft. As the turbine rotates, the generator converts mechanical energy into electrical energy. It is energised 5 seconds after the turbine starts.

3.7 Cooling Pump

The cooling pump circulates cooling water through the condenser to convert exhaust steam back into water (feedwater). It starts after the generator is operational.

3.8 Conveyor

The conveyor system transports fuel (typically coal) from the fuel storage area to the furnace. In the sequence, the conveyor is started after the cooling pump to ensure the fuel supply chain is established after the plant is operational.

3.9 Ash Pump

The ash pump removes bottom ash and fly ash generated during combustion. It is the last component in the sequence, activated after the conveyor is running.

4. PLC and Ladder Logic Fundamentals

4.1 Programmable Logic Controller (PLC)

A PLC is a ruggedised digital computer used for industrial automation. It continuously scans its input modules, executes the control program, and updates its output modules in a cyclic manner known as the scan cycle. PLCs are preferred in industrial environments due to their reliability, ease of programming, and ability to interface with a wide range of sensors and actuators.

4.2 Ladder Logic

Ladder Logic (LD) is a graphical programming language based on the visual appearance of relay logic circuits. A Ladder Logic program consists of rungs, each representing a logical expression. The left and right vertical bars represent power rails. Horizontal rungs contain contacts (inputs) and coils (outputs). When the logical conditions on a rung are satisfied, the output coil is energised.

4.3 Key Instructions Used

The following Ladder Logic instructions are used in this project:

Instruction	Symbol	Description
Examine if Closed (XIC)	— —	Passes power if the referenced bit is 1 (ON)
Examine if Open (XIO)	— / —	Passes power if the referenced bit is 0 (OFF)
Output Energize (OTE)	—()—	Energises the output coil when rung condition is true
On-Delay Timer (TON)	TON Box	Delays turning ON the output by a preset time after input is true
Timer Done Bit (.DN)	T?.DN	Set to 1 when the TON timer reaches its preset value

5. I/O Address Mapping

The PLC uses Allen-Bradley Local I/O addressing conventions. The format is:

Local : Slot : Direction . Data . Bit

Where: Local = Local chassis, Slot = Module slot number, Direction = I (Input) or O (Output),
Data = Data word, Bit = Bit number within the word.

5.1 Input Address Map

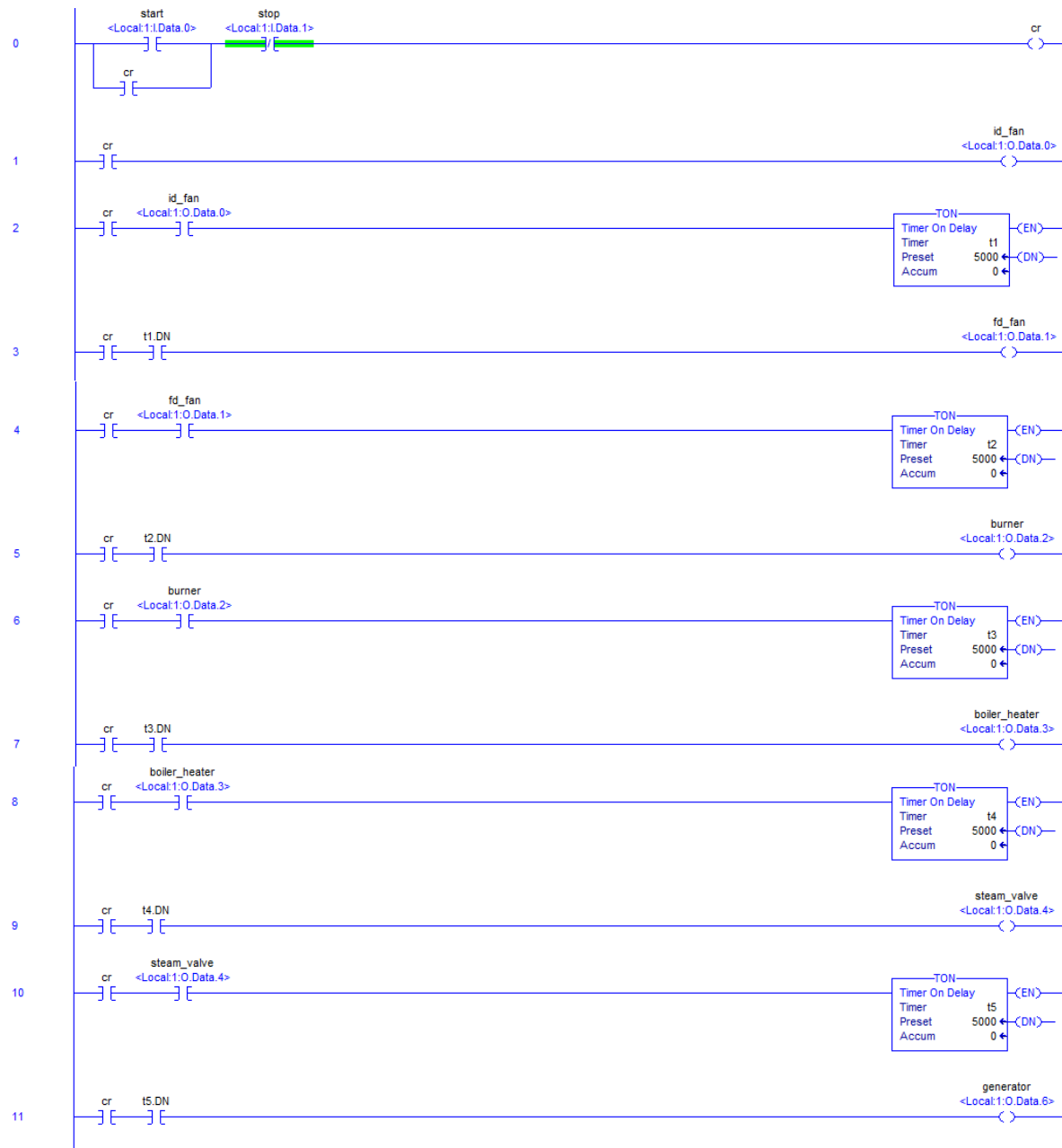
PLC Address	Tag Name	Description
Local:1:I.Data.0	Start	Momentary Start Push Button
Local:1:I.Data.1	Stop	Momentary Stop Push Button (N/C)
Local:2:O.Data.0	id_dan (feedback)	ID Fan running feedback
Local:2:O.Data.1	boiler_htr (feedback)	Boiler Heater ON feedback
Local:1:O.Data.2	bunner (feedback)	Burner ON feedback
Local:2:O.Data.3	boiler_heater (feedback)	Boiler Heater stage 2 feedback
Local:1:O.Data.9	steam_valve (feedback)	Steam Valve open feedback
Local:2:O.Data.6	generator (feedback)	Generator running feedback
Local:3:O.Data.7	cooling_pump (feedback)	Cooling Pump feedback
Local:2:O.Data.0	conveyor (feedback)	Conveyor belt feedback

Block Diagram



6. Ladder Logic Program Description

The ladder logic program for the Thermal Power Plant automation consists of 20 rungs (0 through 19), spanning two pages of handwritten diagrams. Below is a rung-by-rung description of the complete program:





6.1 Rung 0 — Start/Stop and Seal-In Circuit

Rung 0 implements the fundamental Start/Stop control with a seal-in (latch) circuit. The Start contact (Local:1:I.Data.0, XIC) and Stop contact (Local:1:I.Data.1, XIO — normally closed) are in series. A parallel branch with the CR (Control Relay) contact forms the seal-in circuit, keeping the coil energised after the Start button is released. The output energises coil CR.

6.2 Rung 1 — CR to ID Fan Output

When CR is ON (Rung 1), the ID fan output coil (id_dan — Local:2:O.Data.0) is energised. This is the first physical device to start in the plant sequence.

6.3 Rung 2 — ID Fan Timer (t1)

Rung 2 monitors the id_dan coil (Local:2:O.Data.0) and, when the ID fan is confirmed running, activates TON Timer t1 with Preset = 5000 ms (5 seconds), Accum = 0. This introduces a 5-second delay before the next stage.

6.4 Rung 3 — t1.DN to Boiler-Heater Stage 1

When t1.DN (Timer 1 Done Bit) is set, Rung 3 energises the Boiler-Heater output (Local:2:O.Data.1), starting the heating process inside the boiler.

6.5 Rung 4 — Boiler-Heater Timer (t2)

With the Boiler-Heater coil (Local:2:O.Data.1) ON, TON Timer t2 starts counting (Preset = 5000 ms). After 5 seconds, t2.DN is set, triggering the next stage.

6.6 Rung 5 — t2.DN to Burner

When t2.DN is set, the Burner coil (bunner — Local:2:O.Data.2) is energised. This starts combustion in the furnace.

6.7 Rung 6 — Burner Timer (t3)

With the Burner (Local:1:O.Data.2) running, TON Timer t3 starts (Preset = 5000 ms). It ensures the combustion is stable before the boiler heater second stage starts.

6.8 Rung 7 — t3.DN to Boiler-Heater Stage 2

When t3.DN fires, the second Boiler-Heater output (Local:2:O.Data.3) is energised, bringing the boiler to full operational temperature.

6.9 Rung 8 — Boiler-Heater Stage 2 Timer (t4)

With the Boiler-Heater Stage 2 coil (Local:2:O.Data.3) ON, TON Timer t4 starts (Preset = 5000 ms). After 5 seconds, it signals that the boiler has reached operating pressure.

6.10 Rung 9 — t4.DN to Steam Valve

When t4.DN is set, the Steam Valve coil (Local:2:O.Data.4) is energised, allowing high-pressure steam to flow from the boiler towards the turbine.

6.11 Rung 10 — Steam Valve Timer (t5)

With the Steam Valve (Local:1:O.Data.9) open, TON Timer t5 starts (Preset = 5000 ms). This allows pressure to build up in the steam lines before the turbine is engaged.

6.14 Rung 11 — t6.DN to Generator

When t6.DN is set, the Generator coil (Local:2:O.Data.6) is energised. The generator is now mechanically coupled to the turbine and begins producing electrical power.

6.15 Rung 12 — Generator Timer (t7)

With the Generator (Local:1:O.Data.6) running, TON Timer t7 starts (Preset = 5000 ms). This stabilises generator output before the cooling system is started.

6.16 Rung 13 — t7.DN to Cooling Pump

When t7.DN fires, the Cooling Pump coil (Local:2:O.Data.7) is energised. The cooling pump begins circulating cooling water to the condenser.

6.17 Rung 14 — Cooling Pump Timer (t8)

With the Cooling Pump (Local:3:O.Data.7) running, TON Timer t8 starts (Preset = 5000 ms). After 5 seconds, the conveyor belt is started.

6.18 Rung 15 — t8.DN to Conveyor

When t8.DN is set, the Conveyor coil (Local:2:O.Data.0, slot 2) is energised. The conveyor begins transporting fuel to the furnace.

6.19 Rung 16 — Conveyor Timer (t9)

With the Conveyor (Local:2:O.Data.0) running, TON Timer t9 starts (Preset = 5000 ms). After 5 seconds, the ash pump is activated.

6.20 Rung 17 — t9.DN to Ash Pump

When t9.DN fires, the Ash Pump coil (Local:2:O.Data.3, slot 2) is energised. The ash pump removes combustion residue from the furnace, completing the startup sequence.

6.20 Rung 18— CR to Reset t1

When we Force On stop button the CR becomes 0 due to this it resets the t1 timer and we can manually stop the process anytime we want.

7. Timer Configuration

All timers in this project are TON (On-Delay Timer) blocks. Each timer has:

- Timer type: TON (On-Delay Timer)
- Preset value: 5000 ms (5 seconds)
- Accumulator: 0 (starts from zero each time the input goes TRUE)
- Enable bit (EN): Set when the rung condition is true
- Done bit (DN): Set when Accum reaches Preset

Timer	Trigger Input	Preset (ms)	Output (DN triggers)
t1	ID Fan (id_dan)	5000	Boiler-Heater Stage 1
t2	Boiler-Heater 1	5000	Burner (bunner)
t3	Burner	5000	Boiler-Heater Stage 2
t4	Boiler-Heater 2	5000	Steam Valve
t5	Steam Valve	5000	Turbine
t7	Generator	5000	Cooling Pump
t8	Cooling Pump	5000	Conveyor
t9	Conveyor	5000	Ash Pump

8. Working Principle

The control system follows a strictly sequential startup process. When the operator presses the Start button:

- The CN coil is energised and latched through the seal-in contact.
- The ID Fan starts immediately (Rung 1), creating the draft required for safe combustion.
- After 5 seconds (t1.DN), the Boiler-Heater Stage 1 is activated.
- After another 5 seconds (t2.DN), the Burner is ignited, starting combustion.
- After t3.DN, the Boiler-Heater Stage 2 brings the boiler to full temperature.
- After t4.DN, the Steam Valve opens to allow steam flow to the turbine.
- After t5.DN, the Turbine begins rotating under steam power.
- After t7.DN, the Cooling Pump starts to manage condenser cooling.
- After t8.DN, the Conveyor belt starts to supply fuel continuously.
- After t9.DN, the Ash Pump activates to remove combustion residue.

When the Stop button is pressed, the CN coil is de-energised, which immediately de-energises all outputs and stops all running timers, safely shutting down the entire plant.

The total minimum startup time from Start to full operation is $9 \text{ timers} \times 5 \text{ seconds} = 45 \text{ seconds}$ minimum sequential delay, plus the time for each device to physically start.

9. Advantages of PLC-Based Automation

- Safety: Sequential interlocking ensures no component starts before its prerequisites are met.
- Reliability: PLCs are designed for harsh industrial environments with high MTBF.
- Flexibility: The ladder logic program can be modified without rewiring physical circuits.
- Diagnostics: PLC software provides real-time monitoring, fault detection, and alarm handling.
- Reduced human error: Automated sequencing eliminates operator mistakes during startup.
- Energy efficiency: Proper sequencing avoids unnecessary energy consumption during startup surges.
- Scalability: Additional components or stages can be added by extending the ladder program.

10. Conclusion

This project successfully demonstrates the automation of a Thermal Power Plant using PLC Ladder Logic programming. The designed control system implements a safe, sequential startup

sequence for all major sub-systems — from the ID Fan through to the Ash Pump — using nine On-Delay Timers (TON) with a 5-second inter-stage delay.

The ladder logic program ensures that each component is energised only after the previous stage has been confirmed running for the required duration, providing inherent safety interlocking. The use of a seal-in circuit for the Start/Stop logic ensures reliable latching behaviour. The I/O addressing follows standard Allen-Bradley Local conventions, making the program compatible with Rockwell Automation PLC hardware.

This project illustrates the power of PLC-based automation in critical industrial applications where sequential control, safety, and reliability are paramount. The methodology demonstrated here can be extended to more complex power plant control systems incorporating HMI panels, SCADA integration, and fault-tolerant redundancy.

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